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Steganography vs Cryptography: Hidden Security in the Digital World

Presented By

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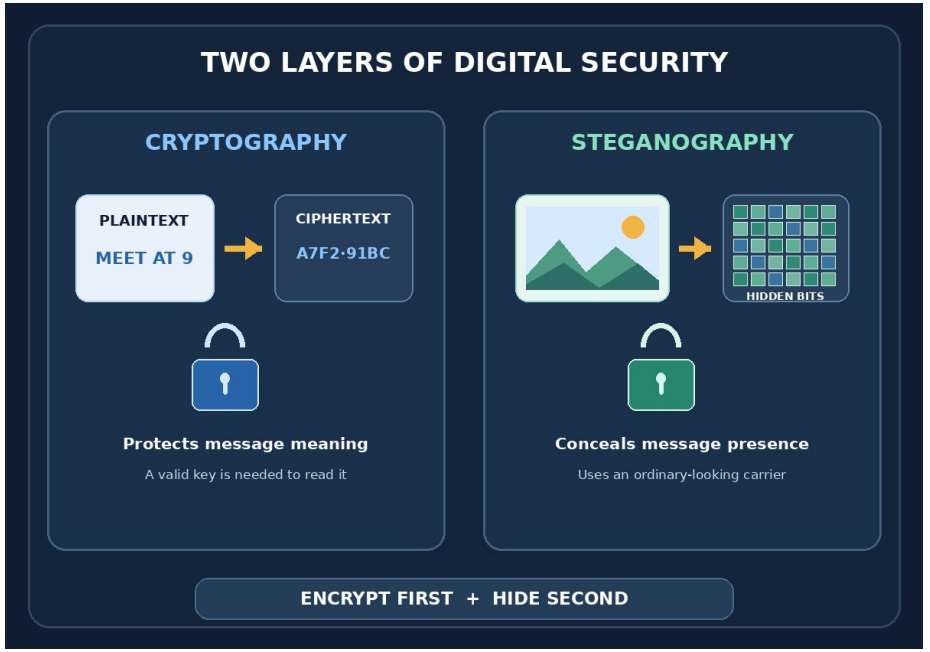
Course: Cryptography and Network Security

ABSTRACT

Digital communication needs methods that protect sensitive information from unauthorized access. Cryptography protects message meaning by transforming plaintext into ciphertext with an algorithm and secret key. Steganography has a different goal: it conceals a message by embedding data in an ordinary digital carrier such as an image. This project compares both methods and demonstrates **an integrated image-based workflow. The** sender provides a message, key, and cover image. The message is encrypted first, then its binary representation is embedded into selected least significant bits (LSBs) of image pixels. The resulting stego image is sent to the receiver, who extracts the hidden payload and decrypts it with the correct key. The workflow includes input validation, encryption and embedding, and extraction and verification. A comparison of security goals, payload capacity, image quality, and key dependence highlights the strengths and limitations of each method. The demonstration shows how encryption protects message content while steganography makes communication less noticeable. Safe key handling, suitable cover files, and testing against image modification remain important for reliable security.

Introduction

- This project compares cryptography and steganography as methods for protecting digital information.
- Cryptography transforms readable data into ciphertext using an algorithm and a secret key, making its content difficult to understand without authorization.
- Steganography hides a message inside an ordinary digital file, such as an image, so that the communication itself is less noticeable.
- The project demonstrates how encryption and image-based Least Significant Bit (LSB) embedding can be combined for layered protection.



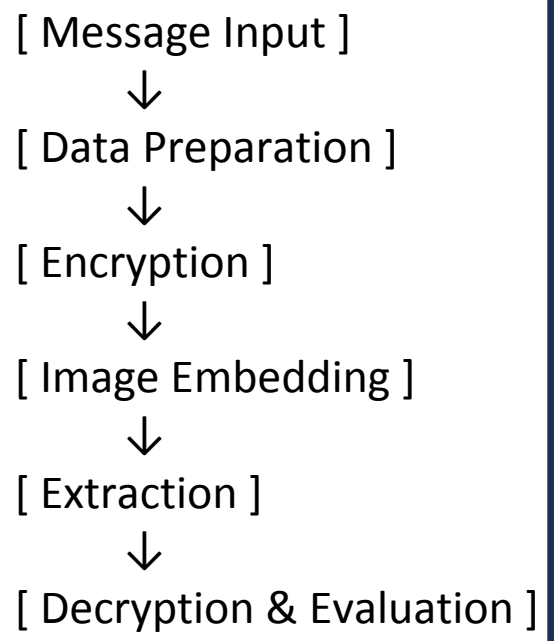
- The comparison highlights confidentiality, concealment, key management, message capacity, and practical limitations.
- The integrated workflow encrypts a message, embeds the ciphertext in a cover image, recovers it at the receiver, and decrypts it. The demonstration shows how an image can carry protected information while ordinary viewing reveals no obvious message.

Objectives

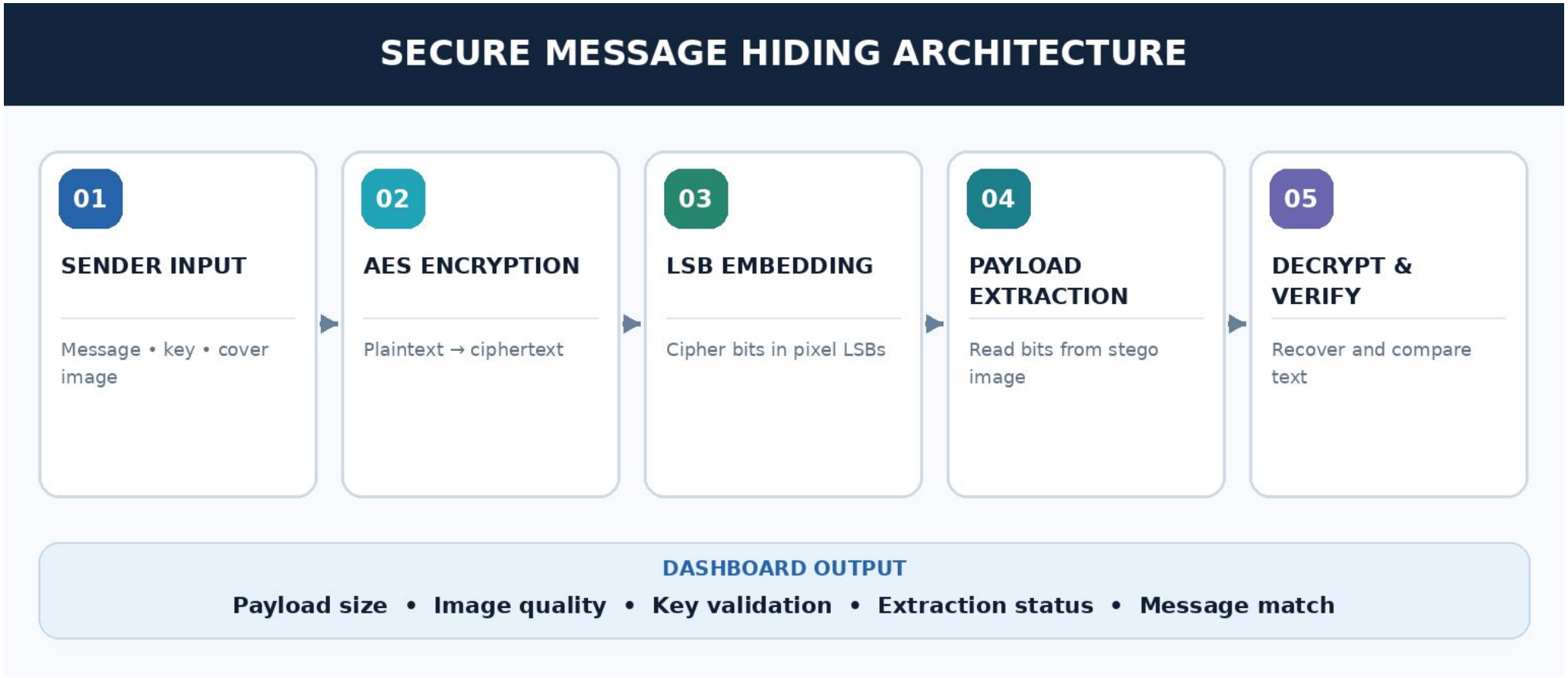
- To analyze cryptography and steganography and explain their roles in secure digital communication.
- To demonstrate how encryption converts plaintext into ciphertext and decryption restores the original message.
- To implement image-based Least Significant Bit (LSB) steganography for embedding and extracting a secret payload.
- To combine encryption and steganography so that a hidden message remains protected if it is discovered or extracted.
- To compare confidentiality, concealment, key dependence, payload capacity, and visual image quality.
- To present the workflow and outputs through clear diagrams, sample results, and an easy-to-read comparison dashboard.

Methodology

- **Message Input:**
  Collect a text message, a secret key, and a suitable cover image from the sender.
- **Data Preparation:**
  Validate the message encoding, image format, file readability, and payload capacity.
- **Encryption:**
  Encrypt the message before hiding it, so extracted payload bits do not reveal the plaintext.
- **Image Embedding:**
  Convert ciphertext into bits and place them in selected pixel least significant bits (LSBs).
- **Message Extraction:**
  Read the embedded bits from the received stego image and rebuild the encrypted payload.
- **Decryption & Evaluation:**
  Use the valid key to recover the message, then verify correctness and inspect image changes.



System Architecture



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MODULE - 1

Message Input & Preprocessing

Message Acquisition:

- Collect the sender's plaintext message, selected cover image, and secret key through the project interface.
- Accept supported image files such as PNG or BMP, validate dimensions, and check whether the encrypted payload fits the available pixel capacity.

Data Preprocessing:

- Normalize message encoding and convert the input into a byte sequence suitable for encryption.
- Validate key format, image readability, file size, payload length, and all mandatory fields before processing.
- Prepare structured input records and pass the validated message, key, and cover image to the encryption and embedding pipeline.

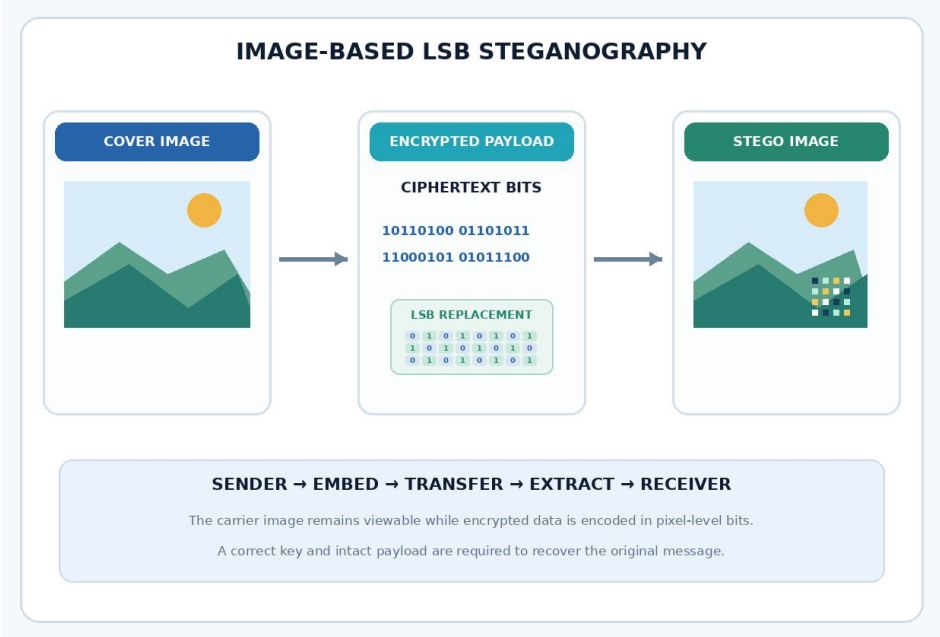
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Steganography Input:

Take input for:

- Secret message or text file
- Encryption key and selected algorithm
- Cover image (PNG/BMP) and payload limit

Encrypt the message, create a binary payload, and embed its bits into the image LSBs. Store the payload length so **the receiver can extract the complete** message reliably.



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MODULE - 2

Implementation using Encryption & LSB Steganography

This stage combines **symmetric encryption** and image-based **LSB steganography** to protect the sender's secret message.

- **Encrypt** the validated plaintext with the selected algorithm and **key**.
- Convert ciphertext into a **binary bitstream** and append a **length marker** for extraction.
- Embed payload bits into selected pixel **least significant bits (LSBs)**.
- **Extract** the bitstream from the received **stego image** and rebuild ciphertext.
- **Decrypt** the recovered ciphertext with the **correct key** to restore the message.
- Compare the recovered message with the **original** and record the **outcome**.

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MODULE - 3

Output and Visualization

Technique Comparison

- Cryptography converts **plaintext into ciphertext; without a valid** key, the message content remains unreadable.
- Steganography conceals the presence **of a message within a cover image,** making the exchange less obvious.
- In **the combined workflow,** ciphertext **is hidden in the image; extraction** alone still does not expose plaintext without the key.
- The comparison graphic summarizes **each technique and shows how layered protection** adds confidentiality and concealment.

AT-A-GLANCE COMPARISON

SECURITY GOAL	CRYPTOGRAPHY	STEGANOGRAPHY	COMBINED
Protects message content	✓ YES	~ NOT ALONE	✓ YES
Hides that a message exists	✗ NO	✓ YES	✓ YES
Uses a secret key	✓ YES	~ OPTIONAL	✓ YES
Main contribution	Confidentiality	Concealment	Both layers

Encryption protects the payload; steganography conceals its presence.

Results (Key Findings)

- Encryption made the **original message** unreadable without the correct key, supporting confidentiality.
- LSB embedding carried the encrypted payload in a cover image with changes **designed to remain visually** subtle.
- Successful extraction and **decryption recovered the original message when the received image and valid key were** used.
- The results demonstrate that cryptography and **steganography serve different goals. Used together, they add confidentiality** and concealment, while capacity, robustness, and key handling remain important limitations.

Future Scope

- **Adaptive Payload & Image Selection:**

  Choose cover images and payload sizes automatically using image dimensions and visual complexity.

- **Stronger Cryptographic Options:**

  Integrate authenticated encryption and safer key-management practices for stronger protection.

- **Robust Steganography:**

  Explore transform-domain methods and test resistance to compression, resizing, and image noise.

- **Security Testing:**

  Add statistical steganalysis checks, tamper detection, and systematic image-quality metrics.

- **Interactive Dashboard:**

  Display embedding capacity, extraction status, key validation, and comparison indicators.

Conclusion

- This project presents an educational comparison of cryptography and steganography for protecting digital communication. It demonstrates encryption and image-based LSB message hiding in a single workflow.
- Cryptography conceals message **meaning, while steganography aims to conceal the existence of the message** itself. The methods therefore address related but distinct security goals.
- In the combined demonstration, ciphertext is embedded in a cover image, recovered from the stego image, and decrypted with the correct key. Successful **recovery verifies the workflow while highlighting dependencies on key** accuracy, image capacity, and correct file handling.
- The project also shows that no single technique removes every risk: encryption strength, key secrecy, payload size, image manipulation, and steganalysis all affect protection. Combining complementary controls can improve defense when accompanied by systematic testing and responsible key management.

Reference

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THANK YOU.